

■■ STEGANOGRAPHY TOOL (HIDE TEXT IN IMAGES)

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Introduction

The Steganography Tool is a simple yet powerful Python project designed to hide secret text messages inside images in such a way that they are invisible to the human eye. This project demonstrates how data can be embedded into digital media without affecting its visual appearance, using the Least Significant Bit (LSB) steganography method.

Abstract

This project focuses on hiding and retrieving secret messages within image files. Using Python along with libraries like Pillow and NumPy, the tool allows a user to encode a message inside an image and decode it later. The project also includes optional password protection through XOR-based encryption, making it secure and interactive for educational and practical applications.

Tools Used

1. Python – Programming language used for development. 2. Pillow (PIL) – For image processing operations. 3. NumPy – For manipulating pixel data at the binary level. 4. Google Colab – To execute and test the project in a cloud-based environment.

Steps Involved in Building the Project

1. Image Upload – The user uploads an image file through Google Colab. 2. Message Encoding – The text message is converted into binary format. 3. Bit Manipulation – The binary bits are embedded into the least significant bits of the image pixels. 4. Image Saving – The modified image is saved as the encoded image. 5. Message Extraction – The process is reversed to retrieve the hidden message. 6. Optional Encryption – The message can be encrypted or decrypted using a password.

Conclusion

This project successfully demonstrates the concept of steganography using simple Python techniques. It helps in understanding how digital data can be hidden within multimedia files without detection. The tool is efficient, lightweight, and serves as a great introduction to data security and image processing concepts.